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# Vision-Based Real-Time Gesture-to-Speech Translation for Sign Language

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**Abstract**

For those who are deaf or hard of hearing, sign language acts as their main means of communication. A gesture-to-speech converter can help close the communication divide between these individuals and the broader community. However, many current approaches come with high computational demands. To address this issue, we propose a convolutional neural network (CNN) architecture optimized for translating sign language gestures to speech, using fewer parameters. Additionally, we integrated the Python library *pyttsx3* for text-to-speech conversion. Our method was tested on two publicly available sign language datasets from America and India, achieving accuracy rates of 98.55% and 99.64%, respectively. The proposed approach also outperforms existing techniques in terms of speed. We further validated its effectiveness through a real-time GUI prototype.

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*Keywords*: Sign Language; Human Computer Interface; CNN; Deep Learning; GUI; Sign Language to speech conversion

## 1. Introduction

Approximately 15% of people worldwide are disabled in some way, with over 466 million people, or more than 5% of the population, experiencing deafness [1]. By 2050, the world's population is projected to nearly triple compared to the year 2000, adding an estimated 500 million individuals. Among them, at least 70 million will face challenges

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related to speech or hearing [2]. These difficulties often create significant barriers to social inclusion, affecting areas such as employment, healthcare, education, and public transportation.

American research that was undertaken revealed that doctors often neglect to teach deaf women essential social skills [3]. Sign language gesture-to-speech interpreters are essential for promoting communication between the deaf and the hearing community [4]. However, training and deploying interpreters in remote or impoverished regions remains a significant challenge [5-6]. This gap highlights the need for accessible support services in all areas, regardless of development status [7]. According to the social services department in Bangladesh, there are 153,776 individuals with hearing impairments, 73,507 with visual impairments, and 9,625 people who have both hearing and visual impairments [8].

Electrochemical devices were initially employed to identify elements of sign language, such as hand positions and angles, prior to its formal recognition in the late 1990s. Early systems relied on large, glove-based devices, which struggled with accuracy and efficiency [9]. Another approach utilized video footage to analyze sign language gestures and generate audio expressions; however, this method faced challenges with frame rates and required manual adjustments for smoother animation [10]. Additionally, the "Intelligent Assistant" [11], which convert speech into text, often produced inaccurate results due to input noise, rendering it ineffective in noisy environments [10]. Compared to these, more advanced systems [12] have been explored, including those employing Bayesian classifiers and radial basis functions for dynamic hand motion recognition. Furthermore, [14] demonstrated that Japanese Sign Language (JSL) could be classified using neural networks (NN). The field of recognition research for Indian Sign Language (ISL) has also made significant progress. One study used a 2D CNN to build a dataset of 26 English letters, achieving 95% accuracy in testing and validation. This study also developed a real-time recognition Graphical User Interface (GUI), facilitating communication for hearing-impaired individuals in India [15].

Bridging the interaction barrier between individuals with speech or hearing impairments and those without remains a significant challenge. Real-time sign language interpretation could greatly improve communication for these individuals. However, many modern sign language recognition techniques suffer from high computational costs, which negatively impact their real-time performance. In this study, we present a real-time sign language translation system that prioritizes computational efficiency, accurately translating sign language gestures into speech using a CNN. Our proposed CNN model is evaluated using two publicly available Kaggle datasets from the U.S. and India.

The key contributions of our work can be outlined as follows:

- Developed a robust CNN model for accurate recognition of sign language.
- Performed a comparative analysis by implementing multiple deep learning techniques.
- Designed and implemented a real-time graphical user interface (GUI) prototype for efficient sign language-to-speech translation.

Section II provides a comprehensive overview of our methodology, while Section III presents the analysis and findings from the experiments. Section IV focuses on the results of real-time GUI predictions. Finally, Section V summarizes our findings and outlines our future work objectives. Within Section III, we detail the experimental setup, dataset descriptions, performance evaluation metrics, results, and comparisons with existing methods.

## 2. Proposed methodology

This section offers a comprehensive explanation of the sign language detection system, covering the layout of the proposed CNN model, as depicted in Fig. 1.

### *2.1. Data Pre-processing*

During this stage, each image in the dataset undergoes several steps to prepare it for model training and evaluation. Firstly, input images are resized to a uniform dimension of 64 by 64 pixels. The dataset is subsequently divided into training and testing sets, with 85% designated for training and 15% reserved for testing. Model training is performed using the training set, while the testing set is reserved to evaluate the model's effectiveness.

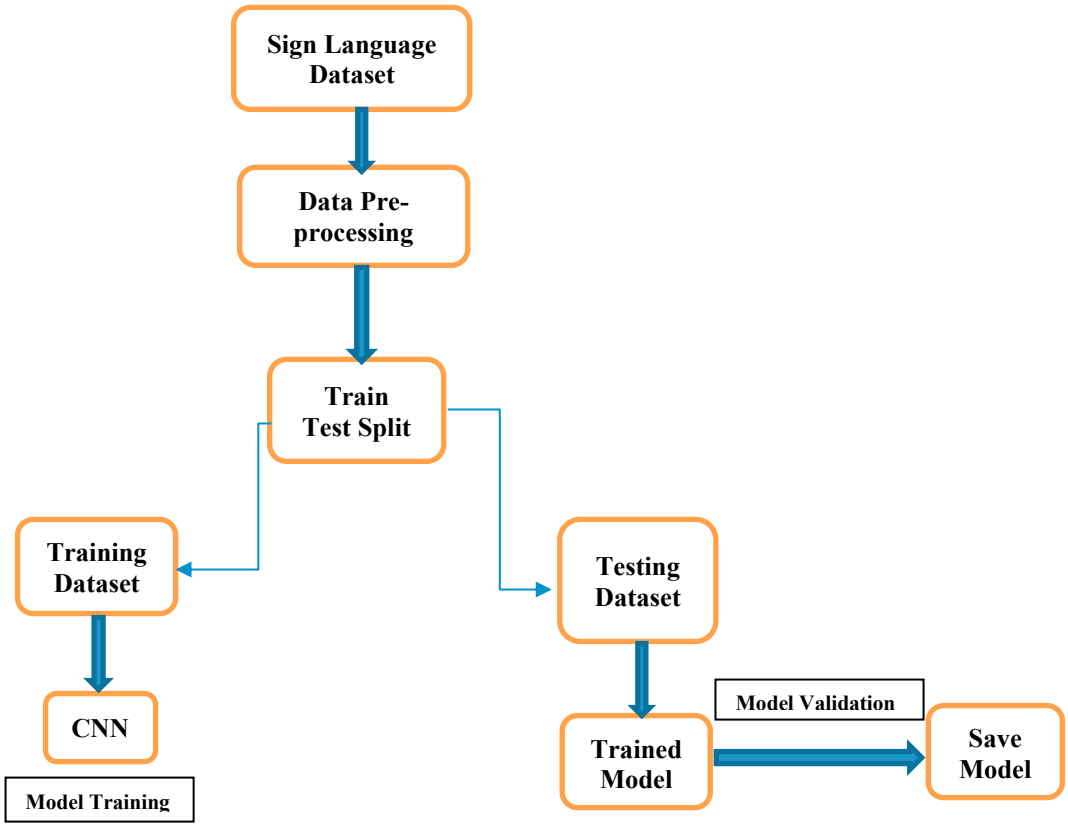



Fig. 1. The sign language detection system's fundamental architectural flow diagram

### 2.2. Model Building

Due to their effectiveness in classification tasks, we utilized convolutional neural networks (CNNs) in this experiment to interpret sign language. The architecture of the CNN is designed to process images with three color channels (RGB) at a size of 64x64 pixels. Details of this architecture are provided in Table 1.

We began by designing a Conv2D layer to process the input data with a shape of 64x64x3, employing 32 filters, each with a 3x3 kernel, along with the ReLU activation function. To reduce the dimensionality of the resulting feature maps, a max pooling layer having a 2x2 pooling size was included. This combination of Conv2D and max pooling layers was repeated once more, using the same parameters, to further enhance feature extraction.

After these layers, we added an additional Conv2D layer having 64 filters, a 3x3 kernel, and ReLU activation to capture more complex patterns. The output was then flattened and passed into a fully connected layer containing 256 neurons. To mitigate overfitting, we incorporated a dropout layer that randomly deactivates 50% of the neurons during training. Finally, to address the multi-class classification task, we incorporated an output layer utilizing a softmax activation function, which is essential for categorizing the input into more than two classes.

### 2.3. Model Training

Table 2 displays the hyper-parameters used for training our proposed CNN model for effective sign language detection task. Using the fit approach we trained the model and following twenty-five epochs, the model showed good accuracy.

Table 1 shows the summary of our proposed CNN model.

| Layer (type) | Output Shape | Param # |
|---|---|---|
| conv2d (Conv2D) | (None, 62, 62, 32) | 896 |
| max_pooling2d (MaxPooling2D) | (None, 31, 31, 32) | 0 |
| conv2d_1 (Conv2D) | (None, 29, 29, 32) | 9,248 |
| max_pooling2d_1 (MaxPooling2D) | (None, 14, 14, 32) | 0 |
| conv2d_2 (Conv2D) | (None, 12, 12, 64) | 18,496 |
| max_pooling2d_2 (MaxPooling2D) | (None, 6, 6, 64) | 0 |
| flatten (Flatten) | (None, 2304) | 0 |
| dense (Dense) | (None, 256) | 590,080 |
| dropout (Dropout) | (None, 256) | 0 |
| dense_1 (Dense) | (None, 35) | 8,995 |

Table 2. Displays the training parameters for our suggested CNN model.

| Parameters | Value | Parameters | Value |
|---|---|---|---|
| Batch-size | 32 | Evaluation - metric | Accuracy |
| Epochs | 25 | Learning - rate | 0.01 |
| Optimizer | Stochastic Gradient Descent (SGD) | Image - size | 64 X 64 |
| Loss Function | Categorical cross entropy | | |

## 3. Experimental results and analysis

This section presents experimental setup, datasets, results, and comparison with existing methods.

### *3.1. Experimental setup*

The proposed CNN algorithm for real-time sign language to speech conversion was implemented in a Jupyter Notebook environment. The system utilized a GPU with RAM 64 GB and a Xeon Gold 6126 x64-based processor by Intel.

We utilized Jupyter Notebook, an offline integrated development environment for Python and machine learning, to train and preprocess our CNN model. The proposed CNN model was tested using a real-time GUI prototype within the Jupyter Notebook environment. For training, we employed two publicly available sign language datasets from Kaggle, one from American named ASL and other from Indian named ISL.

We utilized various Python libraries and machine learning frameworks, including Numpy, Pandas, Keras, and TensorFlow, for dataset preprocessing and model training. Additionally, we used the Python text-to-speech module

PyTTSX3 to transform material written in sign language into spoken English. These frameworks and modules can be installed using the Python pip package manager.

### *3.2. Datasets*

Although numerous sign language databases are available online, most primarily focus on numerical and alphabetical signs in English or other languages. The objective of our system is to convert sign language gestures into English text. To achieve this, we utilized the publicly available ASL and ISL datasets from Kaggle.

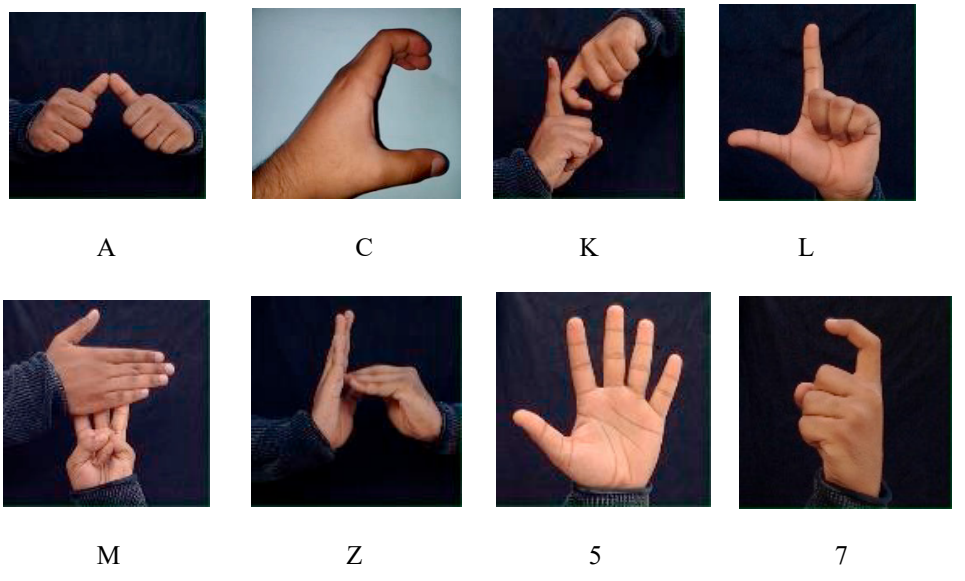



Fig. 2. Sample images of ISL dataset

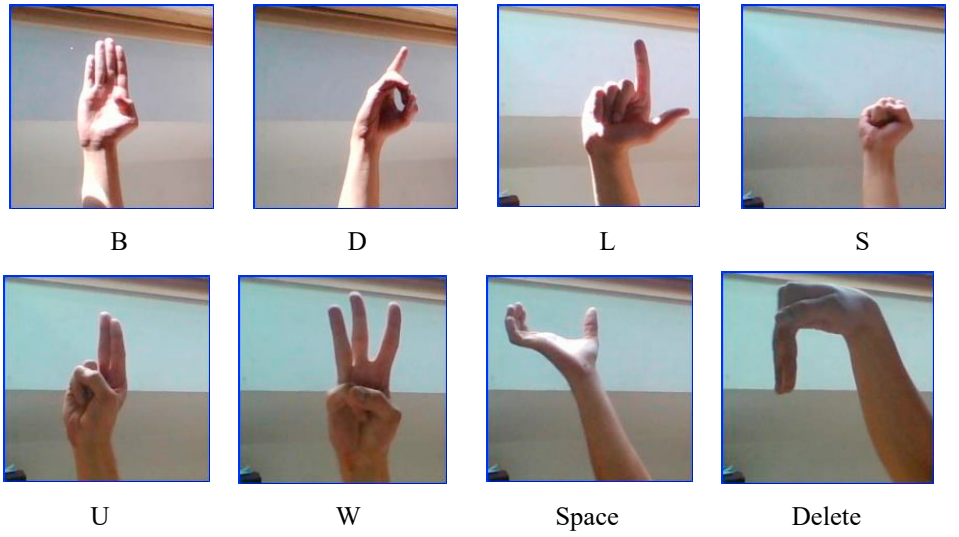


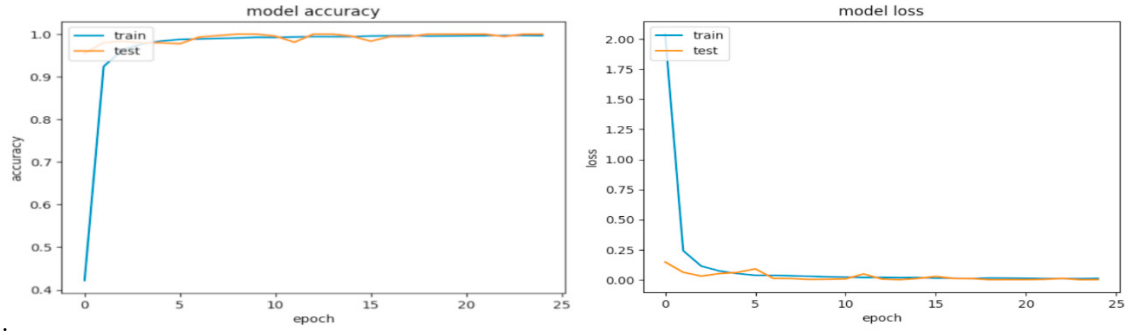

Fig. 3. Sample images of ASL dataset

*ISL dataset:* The ISL dataset [16] comprises 35 classes, encompassing the numerals 1 through 9 and the letters A through Z. It contains a total of 42,745 images, with 1,221 images available for each class. The dataset was created by varying the lighting conditions and poses to ensure robustness in real-time environments. Some sample images are shown in Fig. 2.

*ASL dataset*: The ASL dataset [17] contains 29 classes representing the letters A through Z, as well as SPACE, DELETE, and NOTHING. The collection comprises a total of 87,029 images, with approximately 3,001 images per class. These images were captured in challenging environments to enhance their resilience in real-world situations. Fig. 3 displays a selection of sample images from the datasets used in our study.

### *3.3 Results of ISL dataset*

Accuracy and loss curves offer a clear and concise view of a deep learning model's performance over time. Here, we trained our CNN model over a total of 25 epochs. These curves can be used to detect whether the model is under- or over-fitted. The model trained on the ISL dataset has accuracy and loss curves depicted in Fig. 4. It has been noted that the suggested method successfully classifies the majority of gesture classes. CNN's total accuracy is found to be 99.64%. CNN is therefore operating effectively for the ISL Dataset.




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Fig. 4 shows the accuracy and loss curves of the proposed method on the ISL dataset.

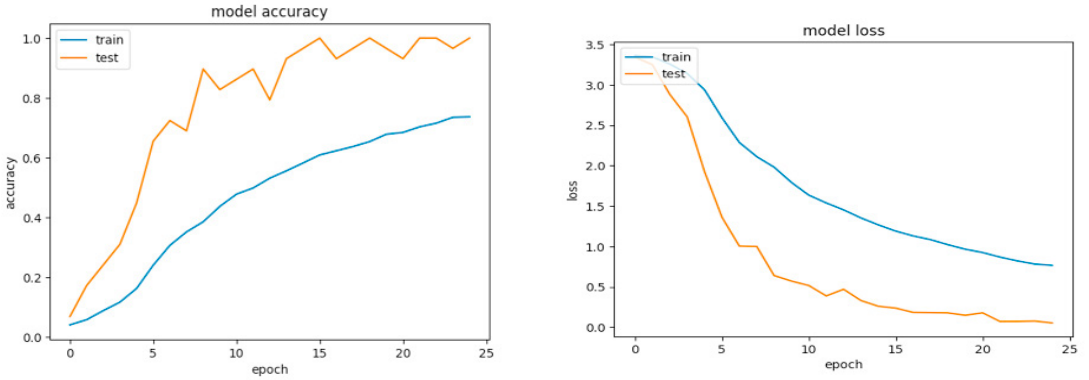



Fig. 5 shows the accuracy and loss curves of the proposed method on the ASL dataset.

### *3.4. Results of ASL dataset*

The performance curves for the model trained on the ASL dataset are presented in Fig. 5. The results indicate that the proposed method successfully classifies the majority of gesture classes. Notably, the CNN achieved an overall accuracy rate of 98.55%, demonstrating its effectiveness on the ASL dataset.

### *3.5. A comparison between the suggested approach and the current approaches*

Table 3 presents the accuracy of the CNN alongside other comparative approaches for both the ISL and ASL datasets. The results show that the CNN achieves a higher accuracy rate compared to other methods. Specifically, on

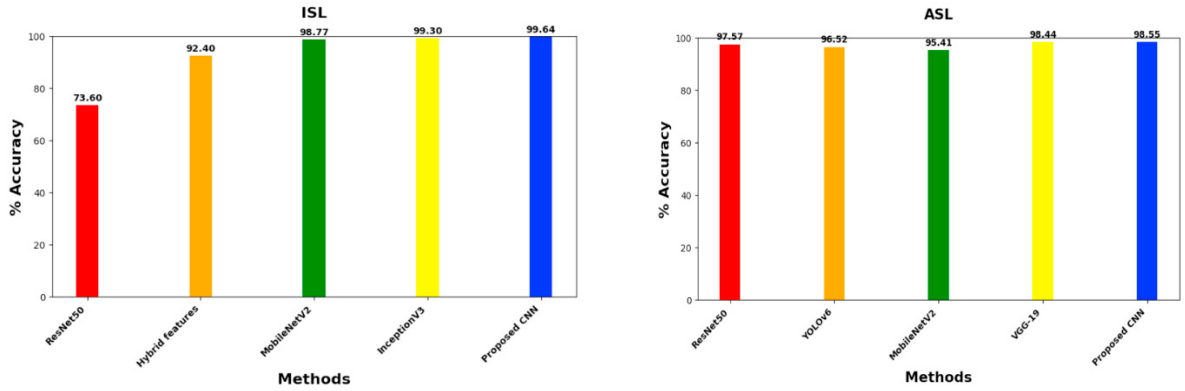
the ISL dataset, the CNN outperformed ResNet50 [21], Hybrid Features [24], MobileNetV2 [20], and mIV3Net [23] by 26.04%, 7.24%, 0.87%, and 0.34%, respectively. Similarly, on the ASL dataset, the proposed CNN demonstrated superior performance, exceeding the accuracy of ResNet50 [19], YOLOv6 [22], MobileNetV2 [18], and VGG-19 [25] by 0.98%, 2.03%, 3.14%, and 0.11%, respectively. The lower classification accuracy of the compared deep learning methods is due to insufficient optimization tailored for this specific task. In contrast, the suggested method's accuracy of recognition is highly competitive, thanks to the optimized number of CNN layers and parameters. Additionally, the inference times of our CNN model are compared with various state-of-the-art methods for effective sign language recognition, as shown in Table 3. Our proposed CNN model exhibits significantly lower computational costs on both ISL and ASL datasets, achieving reduced inference times ranging from 13.62 ms to 124.09 ms. Fig. 9 illustrates real-time predictions made by our trained model using a GUI prototype, demonstrating the model's efficiency and classification accuracy for each sign language gesture.

Table 3. Shows performance comparison of sign language detecting systems.

| Datasets | Method Used | Accuracy (%) | Inference Time (mS) |
|---|---|---|---|
| ISL | ResNet50 [21] | 73.60 | 125 |
| | Hybrid features [24] | 92.4 | 90 |
| | MobileNetV2 [20] | 98.77 | 104.96 |
| | InceptionV3 [23] | 99.3 | 114.32 |
| | Proposed CNN | 99.64 | 76.38 |
| ASL | ResNet50 [19] | 97.57 | 125 |
| | YOLOv6 [22] | 96.52 | 150 |
| | MobileNetV2 [18] | 95.41 | 104.96 |
| | VGG-19 [25] | 98.44 | 200.47 |
| | Proposed CNN | 98.55 | 76.38 |

### 3.6 *Qualitative analysis*

Fig. 6 shows that, in comparison to existing methods for sign language identification, our suggested CNN can more accurately represent significant information.




(a) (b)

Fig. 6. The accuracy on the datasets (a) ISL, (b) ASL

In Fig. 7 class activation map shows that, in comparison to existing methods for sign language identification, our suggested CNN can more accurately represent significant information.

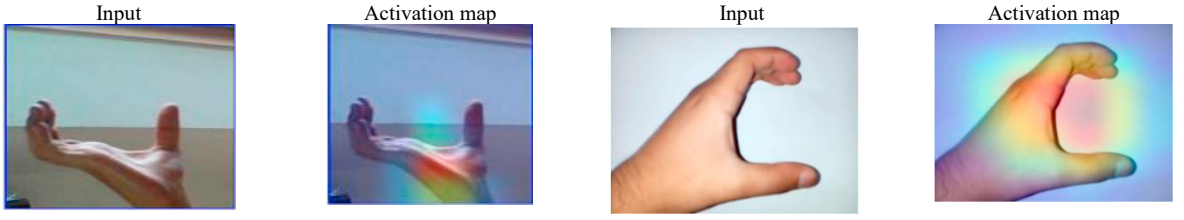



Fig. 7. The class activation map of CNN on ISL and ASL Datasets

## 4. Real-time GUI prototype

We thoroughly train the CNN model using our pre-processed dataset before deploying it for real-time predictions. At this stage, the trained model can recognize and associate signs from the dataset with the provided input sign, predicting the sign that most closely resembles the input. The flowchart of our proposed real-time GUI is depicted in Fig. 8. Additionally, Fig. 9 presents several real-time predictions generated by our CNN model as demonstrated by the GUI prototype. While we conducted extensive experiments with various users, only a selection of sample results is shown.

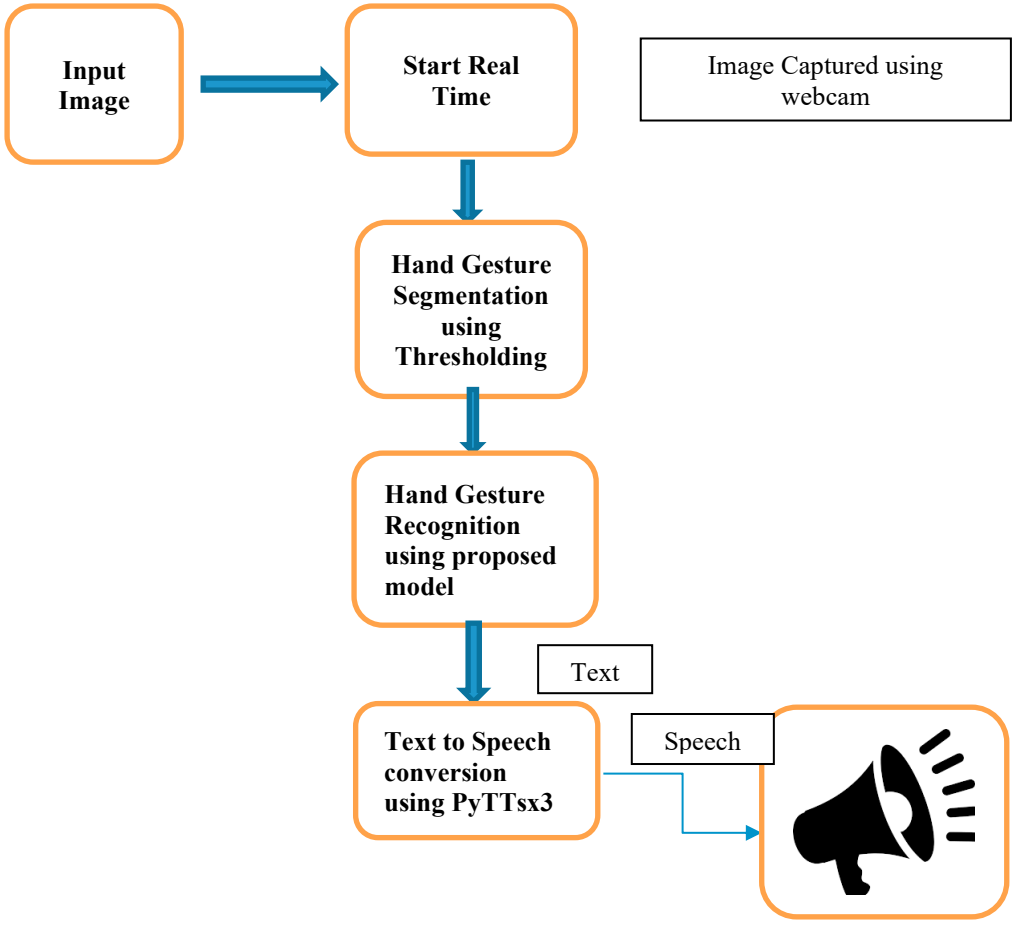
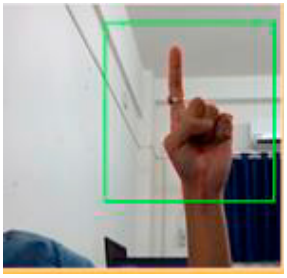


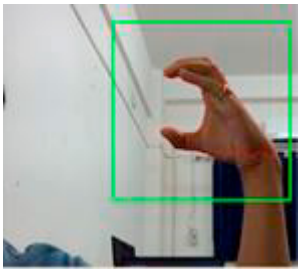

Fig 8. Flow chart for real-time GUI



Input to the GUI prototype




Predicted Output by the GUI



Input to the GUI prototype




Predicted Output by the GUI

Fig 9. Real-time GUI Prototype prediction results

## 5. Conclusions & Future Work

Effective communication with individuals who are deaf requires a robust understanding of sign language. However, one of the primary challenges in developing sign language recognition systems is the high computational cost associated with existing methods. To address this challenge, we designed a CNN-based classifier that efficiently translates sign language movements into speech. This novel approach significantly enhances the recognition of English sign language.

Our model was evaluated using both ISL and ASL datasets, achieving impressive accuracy rates of 98.55% on ASL and 99.64% on ISL. These results represent improvements ranging from 0.34% to 26.64% over state-of-the-art methods. Furthermore, our CNN model significantly reduces the computational burden compared to the methods used in previous studies, with inference times reduced from 124.09 ms to as low as 13.62 ms, making it highly efficient for real-time applications.

The effectiveness of our model is demonstrated through a GUI prototype, which highlights its real-time classification accuracy for various sign language gestures. Moving forward, future research will focus on enhancing accuracy in more complex environments, particularly when distinguishing between similar gestures, to further advance sign language recognition systems.

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We want to extend our deepest thanks to the National Institute of Technology Silchar's Department of Electronics and Communication Engineering for their kind assistance in supplying the necessary materials needed to carry out our research project successfully. Funding for this project was obtained by the SERB CRG (CRG/2022/005850).

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